

To get the latest status for each job using SQL, you can use `JOIN` along with `ROW_NUMBER()` or `MAX(timestamp)`. Below are two approaches:

Schema Example

`job_execution` Table

execution_id	job_name
1	Job_A
2	Job_B
3	Job_A
4	Job_C
5	Job_B

`job_status_history` Table

execution_id	status	timestamp
1	Running	2024-02-01 10:00:00
1	Success	2024-02-01 10:05:00
2	Running	2024-02-01 11:00:00
2	Failed	2024-02-01 11:10:00
3	Running	2024-02-01 12:00:00
3	Success	2024-02-01 12:15:00
4	Running	2024-02-01 13:00:00
5	Running	2024-02-01 14:00:00
5	Success	2024-02-01 14:05:00

Approach 1: Using **ROW_NUMBER()**

This method assigns a ranking to each record per **execution_id** based on the latest **timestamp** and selects only the latest status.

```
WITH LatestStatus AS (  
    SELECT jsh.execution_id, jsh.status, jsh.timestamp,  
           ROW_NUMBER() OVER (PARTITION BY jsh.execution_id ORDER BY jsh.timestamp DESC)  
    FROM job_status_history jsh  
)  
SELECT je.execution_id, je.job_name, ls.status, ls.timestamp  
FROM job_execution je  
JOIN LatestStatus ls ON je.execution_id = ls.execution_id  
WHERE ls.rn = 1;
```

Approach 2: Using **MAX(timestamp)** in a Subquery

```
SELECT je.execution_id, je.job_name, jsh.status, jsh.timestamp  
FROM job_execution je  
JOIN job_status_history jsh  
    ON je.execution_id = jsh.execution_id  
    AND jsh.timestamp = (SELECT MAX(timestamp)  
                        FROM job_status_history  
                        WHERE execution_id = je.execution_id);
```

Expected Output

execution_id	job_name	status	timestamp
1	Job_A	Success	2024-02-01 10:05:00
2	Job_B	Failed	2024-02-01 11:10:00
3	Job_A	Success	2024-02-01 12:15:00
4	Job_C	Running	2024-02-01 13:00:00
5	Job_B	Success	2024-02-01 14:05:00

Which Approach to Use?

- `ROW_NUMBER()` is preferred when you need additional filtering, ranking, or multiple records per execution.
- `MAX(timestamp)` is simpler and works well when you need the latest record.